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A Comparative Approach to Analyzing and Forecasting Carbon Dioxide Emissions in Ethiopia Using Bayesian Autoregressive Integrated Moving Average (ARIMA) and Bayesian Structural Time Series (BSTS) Models

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Abstract

Introduction: Carbon dioxide (CO_2) emissions are a major contributor to global climate change. In Ethiopia, CO_2 emissions have increased steadily in recent decades due to industrialization, rising energy consumption, expanding transportation systems, continued dependence on biomass fuels, and rapid population growth. Although Ethiopia has historically contributed only a small share to global greenhouse gas emissions, the recent upward trend poses challenges for sustainable development and environmental management. Accurate forecasting of CO_2 emissions is essential for designing effective climate mitigation strategies and evidence-based policy decisions. Therefore, this study aimed to analyze and forecast CO_2 emissions in Ethiopia using Bayesian Autoregressive Integrated Moving Average (ARIMA) and Bayesian Structural Time Series (BSTS) models.

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Methods: This study analyzed and forecasted carbon dioxide (CO_2) emissions in Ethiopia using an 82-year time series dataset covering the period from 1941 to 2022. Two Bayesian time series models were employed: Bayesian ARIMA and Bayesian Structural Time Series (BSTS). The Bayesian ARIMA model captured temporal dependencies through autoregressive and moving average components, whereas the BSTS model decomposed the time series into trend, seasonality, and regression components, allowing the incorporation of external predictors. Model parameters were estimated using Markov Chain Monte Carlo (MCMC) simulation techniques. Model performance was assessed using the Watanabe-Akaike Information Criterion (WAIC) and Leave-One-Out Information Criterion (LOOIC), with the model having the lowest values selected as the optimal forecasting model.

Results: The findings revealed that Ethiopia's mean annual per capita CO_2 emissions from 1941 to 2022 were approximately 0.054 metric tons. Among the candidate models evaluated, the Bayesian ARIMA (0, 1, 1) model demonstrated the best fit and forecasting performance based on WAIC and LOOIC criteria. Forecast results from the selected model indicate that Ethiopia's per capita CO_2 emissions are projected to increase gradually, reaching approximately 0.167, 0.169, 0.171, 0.171, 0.175, 0.177, and 0.179 metric tons in 2024, 2025, 2026, 2027, 2028, 2029, and 2030, respectively.

Conclusion: The study indicates a persistent upward trend in Ethiopia's annual per capita CO_2 emissions through 2030, which may intensify climate-related and environmental challenges. These findings underscore the need for strengthened environmental policies and sustainable energy interventions, including carbon taxation, cap-and-trade mechanisms, and the promotion of clean and energy-efficient technologies to reduce future emissions.

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1. Introduction

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Carbon dioxide (CO_2) emissions are the leading contributor to global climate change, posing serious environmental and socioeconomic challenges for both developed and developing countries[1,2]. Climate change, primarily driven by anthropogenic greenhouse gas emissions, has become an increasingly urgent global concern. Among greenhouse gases, CO_2 is the dominant contributor, accounting for more than 75% of total global emissions[2]. Although developed countries contribute the largest share of global emissions, developing countries such as Ethiopia are experiencing a steady increase in CO_2 emissions due to rapid population growth, urbanization, industrialization, and economic transformation [4].

Ethiopia's contribution to global CO_2 emissions is very small in absolute terms; however, its emissions have increased markedly over recent decades. Between 1960 and 2011, CO_2 emissions increased by more than 2000%, mainly driven by fossil fuel combustion and cement production [5]. More recent World Bank data show that Ethiopia emitted 21.1 million metric tons of CO_2 in 2022, representing an increase compared with the previous year [6]. Although the country's absolute emissions remain low globally, the persistent upward trend reflects ongoing structural changes in the economy, including rapid urbanization, industrial expansion, transport growth, and rising energy demand. This dynamic highlights the importance of reliable forecasting approaches to anticipate future emission trajectories and to support evidence-based climate policy and sustainable development planning in Ethiopia.

Despite growing global interest in CO_2 emissions modeling, empirical studies focusing specifically on Ethiopia remain limited. Existing studies have largely concentrated on descriptive trend analysis or conventional econometric approaches, with limited emphasis on forecasting accuracy and uncertainty quantification [7]. This gap restricts the ability of policymakers to design evidence-based climate mitigation strategies, evaluate policy effectiveness,

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and monitor progress toward national and international climate commitments. Accurate CO_2 forecasting is therefore essential for environmental planning, resource allocation, and climate risk assessment [8,9].

To address this gap, this study applies and compares two Bayesian time series models, namely Bayesian Autoregressive Integrated Moving Average (ARIMA) and Bayesian Structural Time Series (BSTS), using annual CO_2 emissions data for Ethiopia from 1941 to 2022[10,11]. These Bayesian approaches are particularly suitable for environmental forecasting because they explicitly quantify uncertainty, allow the incorporation of prior information, and perform effectively under limited or noisy data conditions, which are common in developing-country settings [12,13]. Recent advances in environmental modeling have increasingly emphasized Bayesian frameworks because of their ability to account for uncertainty and structural variability in complex environmental systems. For example, Georgescu and Kinnunen [14] employed Bayesian model averaging to investigate ecological footprint drivers under structural and model uncertainty, while Manigandan et al. [15] compared Bayesian ARIMA and BSTS models for carbon emissions forecasting. Similarly, Bayesian neural network and hierarchical Bayesian approaches have been used to analyze CO_2 emission dynamics and uncertainty under different environmental scenarios [16, 17]. These developments highlight the growing importance of Bayesian methodologies in environmental sustainability and carbon management.

To the best of our knowledge, this study represents one of the first comparative applications of Bayesian ARIMA and BSTS models for forecasting CO_2 emissions in Ethiopia. The novelty of the study extends beyond its geographical focus. Specifically, it provides a comprehensive comparison of two Bayesian time-series approaches, evaluates their predictive performance using Bayesian model selection criteria and

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convergence diagnostics, and quantifies forecasting uncertainty through posterior distributions and credible intervals. Ethiopia provides an important case study because of its ongoing economic transformation characterized by rapid urbanization, industrial expansion, increasing transportation demand, and rising energy consumption, all of which are expected to influence future emissions trajectories. Therefore, the main objective of this study is to identify the model that best captures long-term CO_2 emission dynamics under uncertainty and to provide reliable baseline forecasts that can support climate policy formulation, carbon management, and sustainable development planning. By demonstrating the applicability of Bayesian forecasting approaches in a data-constrained setting, this study contributes to the broader literature on carbon balance, uncertainty quantification, and evidence-based environmental management, thereby aligning closely with the aims and scope of Carbon Balance and Management.

2. Literature Review

Forecasting CO_2 emissions has become increasingly important for countries seeking to balance economic growth with climate commitments and sustainable development goals. Numerous empirical studies have demonstrated a strong relationship between economic growth, energy consumption, industrialization, and CO_2 emissions[17]. Studies conducted in Algeria, Pakistan, and South Africa reported that industrial expansion and increasing energy demand are major drivers of rising emissions [18–20]. However, these relationships vary across countries, with some studies reporting bidirectional causality between economic growth and emissions, while others identify unidirectional effects. In Bangladesh, for example, CO_2 emissions were found to increase more rapidly than both gross domestic product (GDP) and energy consumption, underscoring the urgency of emission control in rapidly developing economies [21]. Despite this growing literature, limited studies have specifically examined Ethiopia, and even

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fewer have applied advanced forecasting methods appropriate for its environmental and data challenges[7].

A wide range of forecasting approaches has been applied in CO_2 emissions research, including classical statistical models, econometric techniques, machine learning algorithms, and hybrid models [22]. Classical methods such as Autoregressive Integrated Moving Average (ARIMA) and Exponential Smoothing (ETS) remain widely used because of their simplicity and interpretability in time series forecasting. However, these methods often rely on strong assumptions regarding stationarity, linearity, and parameter certainty, which may limit their predictive performance under complex environmental conditions [23]. More recently, machine learning models such as Long Short-Term Memory (LSTM), Support Vector Regression (SVR), and hybrid grey models have been used to improve forecasting accuracy in countries such as China, Iran, Malaysia, and Canada [24–28]. For instance, Mustaffa and Shabri developed a grey Bernoulli model optimized using a generalized reduced gradient (GRG) algorithm for forecasting emissions in Malaysia and Singapore, demonstrating improved performance in handling small and noisy datasets[23–25]. Similarly, time series analyses of Malaysia's CO_2 emissions projected continued upward trends in future emissions [25].

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Although these methods have shown promising forecasting performance, many advanced machine learning and hybrid models require large volumes of high-quality data and extensive model tuning, which may not be feasible in low-income countries such as Ethiopia[29]. In addition, many conventional and machine learning approaches provide point forecasts without adequately quantifying uncertainty or accounting for structural changes in the data, limiting their utility for policy applications[30].

Bayesian time series models offer important advantages in addressing these limitations. Bayesian ARIMA extends the classical ARIMA framework by incorporating prior distributions and estimating posterior distributions of

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parameters, thereby improving inference under limited data conditions and enabling probabilistic forecasting [31]. Bayesian Structural Time Series (BSTS) models provide additional flexibility by decomposing time series into trend, seasonal, and regression components while allowing dynamic state evolution[32]. BSTS is particularly useful for capturing structural changes, policy interventions, and external shocks, making it well suited for environmental data characterized by long-term trends and uncertainty.

Despite the recognized advantages of Bayesian approaches, their application in CO_2 emissions forecasting remains limited in Ethiopia and across much of Sub-Saharan Africa[33]. Most previous studies in these settings have relied on traditional econometric or descriptive approaches and have not adequately addressed uncertainty, small sample constraints, or structural shifts in emissions patterns[34] . Furthermore, comparative studies evaluating Bayesian ARIMA and BSTS alongside the broader forecasting literature remain scarce. Therefore, this study contributes to the literature by comparatively applying Bayesian ARIMA and BSTS models to Ethiopia's long-term CO_2 emissions data and assessing their forecasting performance, uncertainty handling, and policy relevance. By doing so, this research provides evidence on the suitability of probabilistic forecasting methods for supporting climate policy and sustainable development planning in Ethiopia.

Recent advances in environmental modelling have increasingly emphasized the use of Bayesian frameworks due to their ability to explicitly account for uncertainty, structural variability, and limited data availability in complex environmental systems. Unlike traditional deterministic approaches, Bayesian methods provide probabilistic inference and support more robust decision-making under uncertainty, which is particularly important in climate and emissions forecasting studies. For example, Georgescu and Kinnunen [35] applied a Bayesian model averaging approach to ecological footprint analysis in Finland, demonstrating how Bayesian techniques can effectively

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integrate model and structural uncertainty and improve the reliability of environmental sustainability assessments. Similarly[36] applied and compared Bayesian ARIMA and BSTS models for forecasting carbon emissions in developing country contexts, highlighting their effectiveness in handling uncertainty in environmental data. In addition, Bayesian neural network and hierarchical Bayesian approaches have been employed to analyze CO_2 emission dynamics and environmental determinants in China and Africa, providing robust inference under conditions of data limitations and structural uncertainty[37,38]. Collectively, these studies highlight the growing importance of Bayesian methodologies in environmental forecasting and strongly support their application in this study for modeling and forecasting Ethiopia's CO_2 emissions under uncertainty and limited data availability.

3. Methods and Materials

3.1. Description of the Study Area (Revised)

This study focuses on Ethiopia, a landlocked country located in the Horn of Africa. Ethiopia is characterized by diverse topography, including highland plateaus, lowlands, and the Great Rift Valley, which contribute to variations in climate, land use, and economic activity across regions[39]. The country has a rapidly growing population and is undergoing significant structural transformation driven by urbanization, industrial development, expansion of infrastructure, and increasing energy demand. These factors are closely linked to changes in energy consumption patterns and greenhouse gas emissions. Ethiopia also plays an important geopolitical role in Africa and hosts major international institutions, including the African Union headquarters in Addis Ababa. These socioeconomic and developmental characteristics make Ethiopia a relevant case for analyzing long-term carbon dioxide emission trends and forecasting future trajectories[40].

3.2. Data Source

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This study used secondary data obtained from national emission databases to analyze carbon dioxide (CO_2) emissions in Ethiopia. The dataset comprises 82 annual observations from 1941 to 2022, sourced from the Our World in *Data* platform and national statistical agencies. The data are expressed in metric tons per capita. For reference, the official World Bank indicator can be accessed at: https://databank.worldbank.org/databases [41]. CO_2 emissions are the principal greenhouse gas contributing to climate change, primarily resulting from fossil fuel combustion, electricity generation, and cement production. The volume of CO_2 emitted varies according to the type and consumption level of fossil fuels.

3.3. Method of Data Analysis

3.3.1. Data Preparation

The dataset was reviewed for missing values and anomalies. All inconsistencies were carefully addressed to preserve the integrity of the data. Statistical analyses were conducted using R software, employing the rstan package for Bayesian inference and the bsts package for Bayesian Structural Time Series modelling.

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3.3.2. Summary Statistics

Initial descriptive statistics including the mean, minimum, maximum, standard deviation, skewness, kurtosis, and the Jarque-Bera normality test were used to understand the distributional properties of the CO_2 emission data. Correlation analysis was conducted to explore relationships between CO_2 emissions and other variables.

3.3.3. Stationarity Testing

Stationarity, which implies constant statistical properties over time, is essential for reliable time series forecasting. A non-stationary series can result in misleading trends in ARIMA models[42]. To test for stationarity, the Augmented Dickey–Fuller (ADF) test was applied. If the data exhibited non-

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stationarity, first-order differencing was performed. If needed, a second difference was applied to achieve stationarity. This transformation helps ensure accurate model estimation and forecasting.

3.3.4. ACF and PACF Analysis for Model Identification

Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF**)** plots were employed to identify significant lags in the time series and guide the selection of ARIMA parameters. These plots informed the specification of the model's **(p, d, q)** values, where p is the order of the autoregressive term, **d** is the differencing level, and ***q*** is the order of the moving average component [43].

3.3.5. Bayesian Analysis of ARIMA and BSTS Models

Bayesian inference provides a structured way to incorporate prior knowledge with observed data to estimate model parameters and quantify uncertainty. In this study, both Bayesian ARIMA and Bayesian Structural Time Series (BSTS) models were implemented using the Hamiltonian Monte Carlo (HMC) algorithm via RStan [11]. Model evaluation was conducted using the Leave-One-Out Information Criterion (LOOIC) and Watanabe-Akaike Information Criterion (WAIC)[44]. The model with the lowest LOOIC and WAIC values was selected as the best fit for forecasting[45]. The Bayesian ARIMA framework is well-suited to address complexities such as non-stationarity, multiple trends, and model uncertainty. It allows for the incorporation of informative priors and supports robust model comparison and selection.

3.3.5.1. Model Identification and Estimation

Initial model selection involved visual inspection of time series plots and ACF/PACF analysis. The ADF test was used to assess stationarity. Differencing was applied if necessary, taking care to avoid over-differencing. Model parameters were estimated and refined using residual diagnostics. If

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model inadequacy was found, adjustments were made, and the process was repeated. Final model selection was based on a combination of diagnostic fit and predictive performance metrics (see Table 1).

Table 1: Behavior of theoretical ACF and PACF Patterns Bayesian ARIMA Model (see in Result figure 6)

Model	ACF	PACF
AR(p)	Dies down/exponential decay	Cut off after lag q
MA(q)	Cut off after lag q	Dies down/exponential decay
ARMA(p,q)	Dies down/exponential decay	Dies down/exponential decay

Source: *Time series analysis and its application with R [42]*

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3.3.6. Bayesian Structural Time Series

The Bayesian Structural Time Series (BSTS) model analyzes variability in time series by estimating observation variance (s_obs), level variance (s_level), and slope variance (s_slope). Trace plots assess model convergence, with inconsistent chains indicating potential issues [11]. Autocorrelation plots reveal correlations between observations at different lags, with values near 1 indicating strong positive correlation. Histograms show a positively skewed distribution for observation variance, while density plots suggest multiple plausible parameter values (multimodality). Caterpillar plots display credible intervals, where wider intervals indicate more uncertainty. Overall, s_obs reflects varying observation noise, whereas s_level indicates a more stable underlying trend.

Given a sequence of observations $\underline{y}: y_1, y_2, \cdots, y_t$ from a strictly stationary and invertible ARMA model 1, the task is to formulate the approximate likelihood function for an ARMA (p,q) model with y_t's representing the observed time series data.

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$$y_t \sim N\left(\left(\mu_0 + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}\right), \sigma^2\right)$$

Under the assumption of $y_t = \epsilon_t = 0$ for $t \leq 0$. The condional density of y_t given y_{t-1} , y_{t-2} , $\cdots$, y_{t-p} can be written as [11],

$$f(y_{t-1}, y_{t-2}, \cdots, y_{t-p} ; \mu_0, \Phi, \Theta) \propto$$

$$\left(\frac{1}{\sigma^2}\right) \exp\left(-\frac{1}{2\sigma^2}\left(y_t - \mu_0 - \sum_{i=1}^{p} \phi_i y_{t-i} - \sum_{j=1}^{q} \theta_j \epsilon_{t-j}\right)^2\right)$$

The likelihood function is defined as:

$$f(\underline{y} \mid \mu_0, \Phi, \Theta) \propto \prod_{t=p+1}^{T} f(y_t \mid y_{t-1}, y_{t-2}, \cdots, y_{t-p} ; \mu_0, \Phi, \Theta)$$

Which reduces to:

$$f(\underline{y} \mid \mu_0, \Phi, \Theta) \propto \left(\frac{1}{\sigma^2}\right)^{(T-p)/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{t=p+1}^{T}\left(y_t - \mu_0 - \sum_{i=1}^{p} \phi_i y_{t-i} - \sum_{j=1}^{q} \theta_j \epsilon_{t-j}\right)^2\right)$$

3.3.6.1. Local Level State Component

The local level model represents the most basic form of a structural time series model. In this model, it is presumed that the trend behaves as a random walk[47]

$$y_t = \alpha_t + \epsilon_t \qquad \epsilon_t \sim N\left(0, \sigma_\epsilon^2\right)$$

$$\alpha_{t+1} = \alpha_t + \eta_t \qquad \eta_t \sim N\left(0, \sigma_\epsilon^2\right)$$

In the local level model, the matrices Z_t , T_t, and R_t in the equation are simplified to a single scalar value of '1'. The model's parameters consist of the variances of the error terms$\left(\sigma_\epsilon^2, \sigma_\eta^2\right)$

3.3.6.2. Local Linear Trend State Component

In the local linear trend model, it is assumed that both the average and the slope exhibit random walk behaviour. The formula for the average is as follows:

$$y_t = \mu_t + \epsilon_t \qquad \epsilon_t \sim N\left(0, \sigma_\epsilon^2\right)$$

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$$\mu_{t+1} = \mu_t + \delta_t + \eta_t \qquad \eta_t \sim N\left(0, \sigma_\eta^2\right)$$

The slope equation is:

$$\delta_{t+1} = \delta_t + \zeta_t \qquad \zeta_t \sim N\left(0, \sigma_\zeta^2\right)$$

The local linear trend model is frequently chosen for modelling trends due to its rapid adjustment to local variations. It is particularly useful for short-term forecasting. However, for longer-term predictions, this adaptability may not be ideal, as it often leads to excessively wide uncertainty ranges in forecasts[48]

3.3.7. Parameter Estimation for Bayesian ARIMA and BSTS Models

The ARIMA model is characterized by three parameters (p, d, q), representing the autoregressive order, differencing degree, and moving average order, respectively. After selecting the appropriate model, parameters for the AR and MA components are estimated using Maximum Likelihood Estimation (MLE), which finds values maximizing the likelihood of the observed data. Significance tests and standard errors accompany these estimates. In R, MLE for Bayesian ARIMA aligns with minimizing the sum of squared errors[11]. For the Bayesian Structural Time Series (BSTS) model, parameter estimation begins by specifying prior distributions for error variances, typically inverse Gamma distributions assumed independent across model equations. Posterior distributions are then derived using Markov Chain Monte Carlo (MCMC) simulation techniques, implemented via the BSTS package in R developed by Steven L. Scott [49]. Filtering updates the model as new data arrives, reflecting the Markovian property where each state depends only on the previous one. Forecasts derive from the posterior predictive distribution, with MCMC enabling robust model fitting.

3.3.8. Forecasting and Accuracy Measures of CO_2 Emissions

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After selecting the optimal Bayesian ARIMA model, forecasts of CO_2 emissions are generated along with confidence intervals[50]. These forecasts offer valuable insights for informing environmental policy. Model accuracy is evaluated by comparing predicted values with actual emissions data as it becomes available. Forecast performance is quantified using metrics such as the Mean Absolute Percentage Error (MAPE), where a value around 10% indicates strong predictive accuracy. Significant deviations between forecasts and observations may suggest the need to incorporate additional variables like economic factors or policy changes in future models.

3.3.9. Model Selection Criteria

To identify the best-fitting model, this study applies Bayesian model selection metrics including the Leave-One-Out Information Criterion (LOOIC) and Widely Applicable Information Criterion (WAIC) [51]. Both LOO-CV and WAIC estimate out-of-sample prediction accuracy by evaluating log-likelihoods from posterior parameter simulations. These criteria provide advantages over traditional metrics such as AIC and DIC by better accounting for model complexity and predictive uncertainty.

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4. Result and Discussion

4.1. Descriptive Statistics for CO_2 Emissions in Ethiopia

The results in Table 2, Ethiopia's average annual CO2 emissions from 1941 to 2022 stood at 0.054. The skewness value of 0.985 suggests a positive skew in the data, indicating a lack of symmetry. A kurtosis value of 0.597 points towards a platykurtic distribution. These findings align with the results of the Jarque-Bera (JB) test, which rejects the null hypothesis that carbon dioxide measurements follow a normal distribution. The JB test results (with a JB value of 4.908) indicate that the observed data series does not exhibit a normal distribution.

Table 2: Descriptive Statistics for CO_2 Emissions

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Description	CO_2 Emissions in Ethiopia
Mean	0.054
Median	0.052
1st Quartile	0.020
Skew	0.985
Kurtosis	0.597
Min	0.000
Max	0.157
Range	0.157
3rd Quartile	0.064
Standard Deviation	0.040
Jarque Bera	4.908
Probability	0.086

4.2. Test of Stationary Nature of the Dat

The initial step in time series analysis is to examine the graphical behavior of the observed data over time. Figure 1 presents the annual carbon dioxide (CO_2) emissions per capita in Ethiopia from 1941 to 2022. The plot indicates an overall upward trend with fluctuations over time, suggesting that the series does not exhibit a constant mean and may be non-stationary.

To formally test this, the Augmented Dickey–Fuller (ADF) **test** was applied to the original CO_2 emissions series. The ADF test examines the presence of a unit root, which is an indicator of non-stationarity in time series data. The results in Table 3 show a test statistic of -0.87991 with a p-value of 0.9508, which is greater than the 0.05 significance level. Therefore, the null hypothesis of a unit root cannot be rejected, confirming that the original series is non-stationary. To address this issue, first-order differencing (d = 1**)** was applied to remove the trend component and stabilize the mean of the series. The differenced series was then reassessed to ensure stationarity. The results confirmed that the transformed series was stationary. In addition, the ACF and PACF plots (Figures 2 and 3) were generated using the first-differenced series. These plots show no strong or persistent autocorrelation patterns, suggesting that the differenced series is adequately stationary and

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that only a simple autoregressive/moving average structure is required. Based on these results, the CO_2 emissions series is modeled using a Bayesian ARIMA specification with integration order d = 1, corresponding to a Bayesian ARIMA (0,1,1) model for the original series.

Table 3: Results of the Augmented Dickey-Fuller (ADF) Test on Original Carbon Dioxide Emissions in Ethiopia

ADF	Dickey-Fuller	Lag order	p-value
Value	-0.87991	4	0.9508

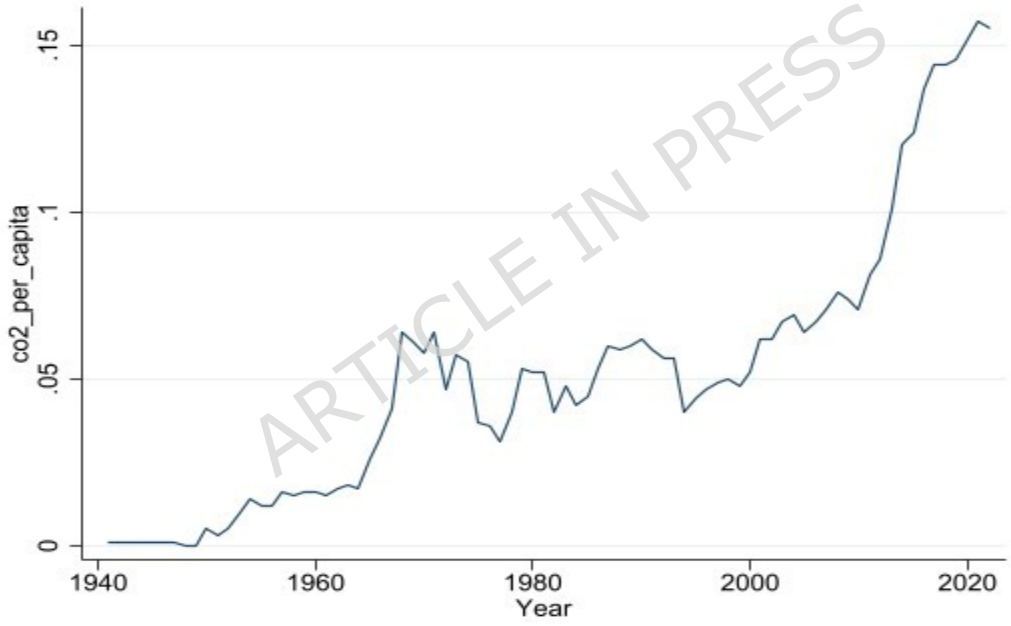


Figure 1: Time Series Plot of CO_2 Emissions in Ethiopia from 1941 to 2022

The carbon dioxide emissions series was stabilized after first differencing (d=1). The Augmented Dickey-Fuller (ADF) test confirmed that the first-differenced data is stationary, which is also supported by the ACF and PACF plots of the differenced series shown in Figure 2 and 3.

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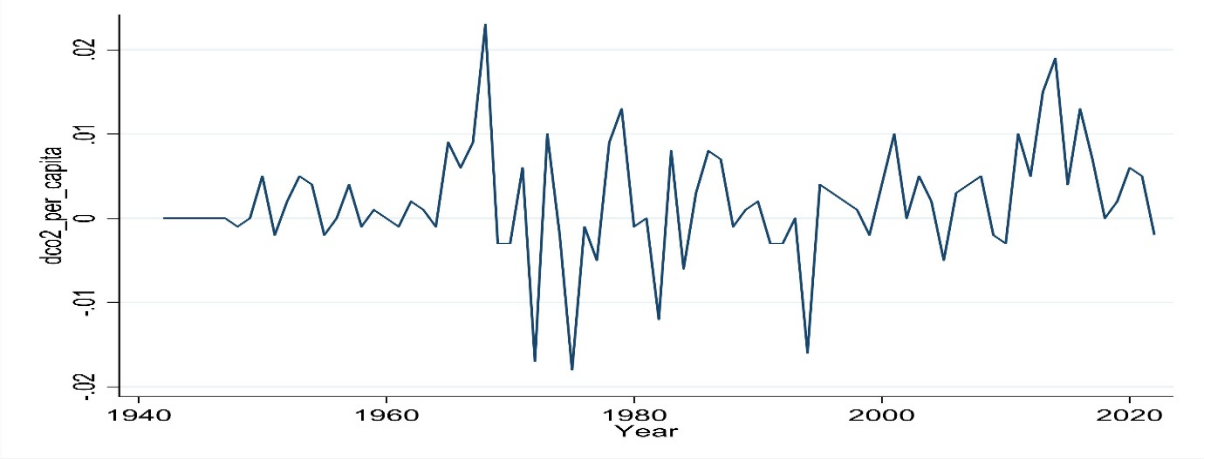


Figure 2: Trend plot of the first-differenced CO_2 emissions in Ethiopia.

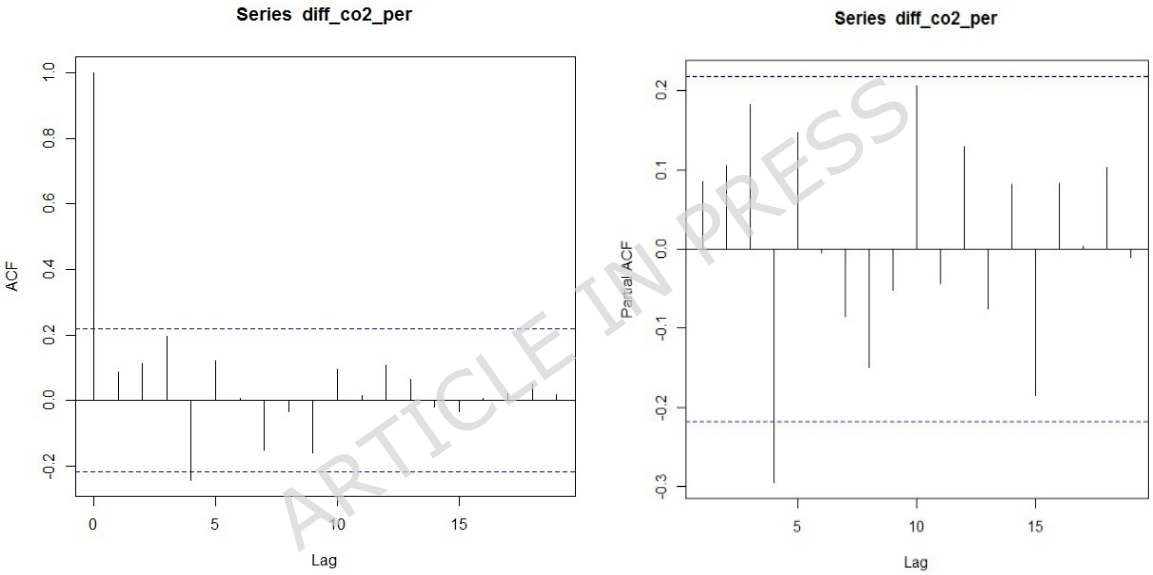


Figure 3**:** ACF, and PACF plots of the first-differenced CO_2 emissions in Ethiopia.

The modelling process can begin once the Augmented Dickey-Fuller (ADF) test confirms that the CO_2 emissions data is stationary after first differencing. This indicates that an ARIMA (p, 1, q) model is appropriate for the differenced time series, as shown in Table 4

Table 4: Dickey-Fuller Test for First Difference CO_2 Emissions Data

Difference	Dickey-Fuller	Lag order	P-value
1st	-3.6247	4	0.03657

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4.3. Analysis of Autoregressive Integrated Moving Average (ARIMA) model

In this study, model selection for the Bayesian ARIMA models was conducted using two fully Bayesian model comparison criteria: the Widely Applicable Information Criterion (WAIC) and the Leave-One-Out Information Criterion (LOOIC)**.** These measures are derived from the posterior predictive distribution and evaluate out-of-sample predictive performance while accounting for model uncertainty. In both criteria, lower values indicate better predictive accuracy. Unlike classical information criteria (e.g., AIC or BIC), WAIC and LOOIC are based on the full posterior distribution obtained via Markov Chain Monte Carlo (MCMC) sampling, making them more appropriate for Bayesian model comparison. As shown in Table 5, the Bayesian ARIMA (0,1,1) model achieved the lowest WAIC (-546.3) and LOOIC (-546.2), indicating superior predictive performance compared to the alternative candidate models. The differences in expected log predictive density (ELPD) among competing models were relatively small, suggesting that although several models perform similarly, ARIMA (0,1,1) provides the best balance between fit and predictive accuracy. Accordingly, the Bayesian ARIMA (0,1,1) model was selected as the optimal model for forecasting CO_2 emissions in Ethiopia.

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Table 5: Model Selection Results for Bayesian ARIMA Models of CO_2 Emissions in Ethiopia

Bayesian ARIMA Model	**WAIC**	**LOOIC**
ARIMA(2,1,2)	-544.3	-543.9
ARIMA(0,1,0)	-542.2	-542.1

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ARIMA(1,1,0)	-544.6	-544.6
ARIMA(0,1,1)	**-546.3**	**-546.2**
ARIMA(1,1,1)	-544.8	-544.2

4.3.1. Posterior Estimates for the Bayesian ARIMA (0,1,1) Model

Posterior inference for the Bayesian ARIMA (0,1,1) model was obtained using Markov Chain Monte Carlo (MCMC) simulation. The model was run with multiple chains and sufficient iterations to ensure convergence and stable posterior estimates. Table 6 summarizes the posterior distribution results, including the posterior mean, standard deviation, quantiles, effective sample size (n_eff), standard error of the mean (Se_mean), and the Gelman–Rubin convergence diagnostic ($\hat{R}$). The results in Table 6 indicate that the effective sample sizes are sufficiently large (n_eff > 100 for all parameters), suggesting adequate sampling efficiency. In addition, all $\hat{R}$ values are approximately equal to 1, confirming that the MCMC chains have converged satisfactorily. The parameter μ represents the posterior mean of the differenced series, while σ represents the innovation standard deviation. The narrow credible intervals and stable convergence diagnostics indicate that the posterior estimates are reliable for forecasting purposes.

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Table 6: **Posterior Summary Statistics for the Bayesian ARIMA (0, 1, 1) Model**

Parameters	**Mean**	**Se_mean**	**Sd**	**2.5%**	**50%**	**97.5%**	**n_eff**	**Rhat($\hat{R}$)**
mu	0.002	0	0.001	0.001	0.002	0.003	9511	1.000

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Sigma	0.007	0	0	0.006	0.007	0.008	2362	1.001

For each parameter, n_eff *provides an estimate of the effective sample size, indicating the number of independent draws from the posterior. The Gelman–Rubin diagnostic statistic (*$\hat{R}$*) represents the potential scale reduction factor; values close to 1 suggest that the MCMC chains have converged adequately.*

A trace plot, as shown in Figure 4, provides a visual assessment of the convergence of the Markov chains by plotting simulated parameter values across iterations. This diagnostic complements the Gelman-Rubin statistic ($\hat{R}$) in evaluating whether the chains have mixed well. In the Bayesian ARIMA (0, 1, 1) model, the trace plot displays multiple chains for each parameter, and their overlapping behavior suggests good mixing and convergence to the posterior distribution. In contrast, the caterpillar plot in Figure 5 displays the 95% credible intervals for each model coefficient. Although the intervals are well-defined, the fact that they cross the zero line indicates that none of the covariates are statistically significant. Thus, while the model demonstrates convergence, individual parameters may not have strong explanatory power based on the current data.

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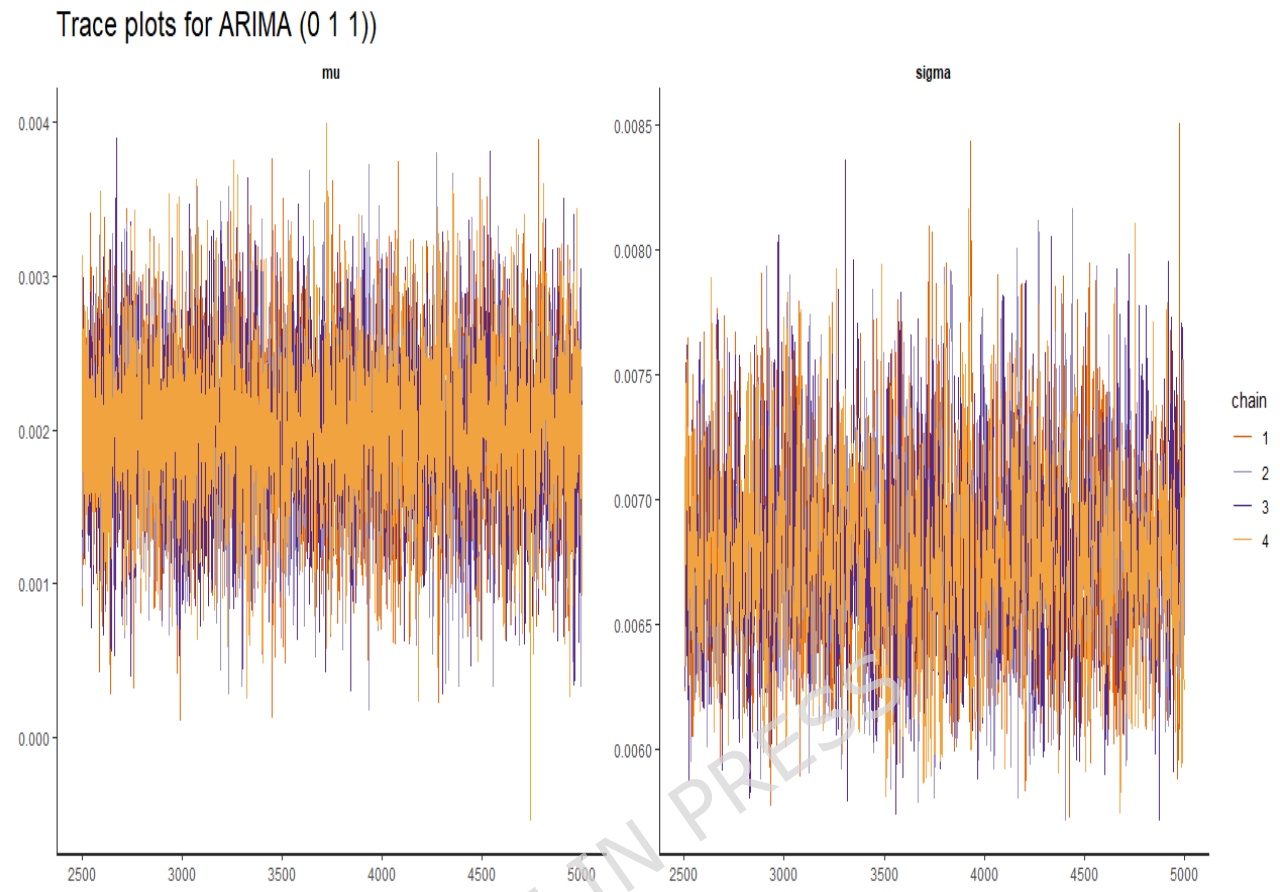
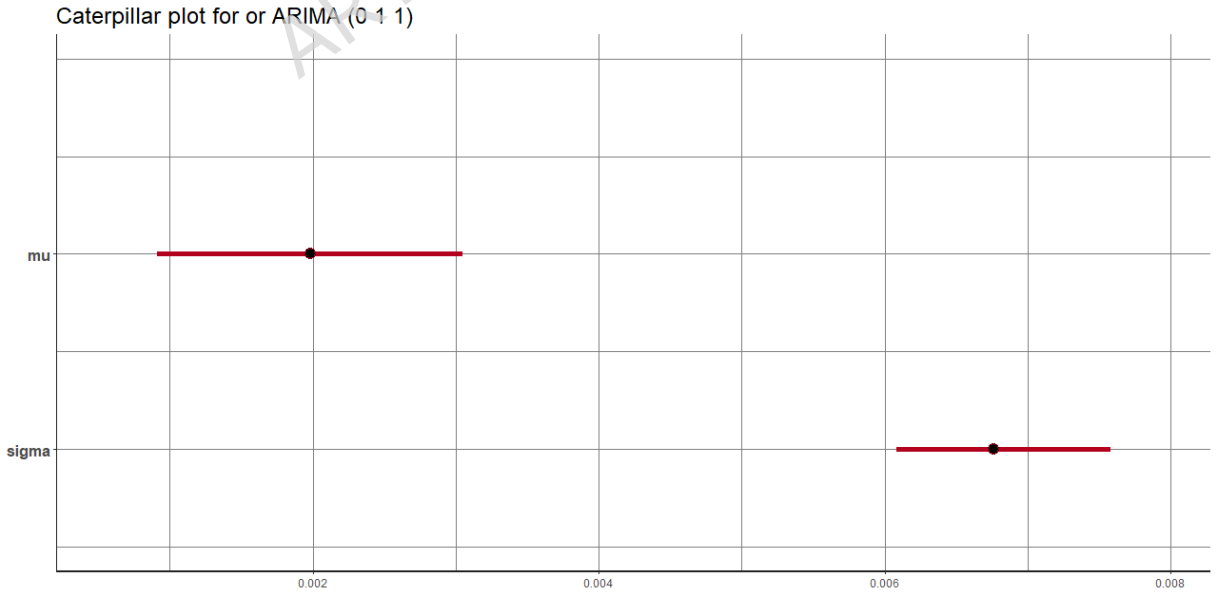


Figure 4: Trace plots of the posterior distributions for the Bayesian ARIMA (0, 1, 1) model parameters.



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Figure 5: Trace plots of the posterior distributions for the Bayesian ARIMA (0, 1, 1) model parameters

Figure 6 presents a posterior predictive density (PPD) plot, where the density of the observed values (*y*) is overlaid with the densities of the replicated values (*y rep*) in varying shades of blue. The visual alignment between these distributions provides evidence of a good model fit.

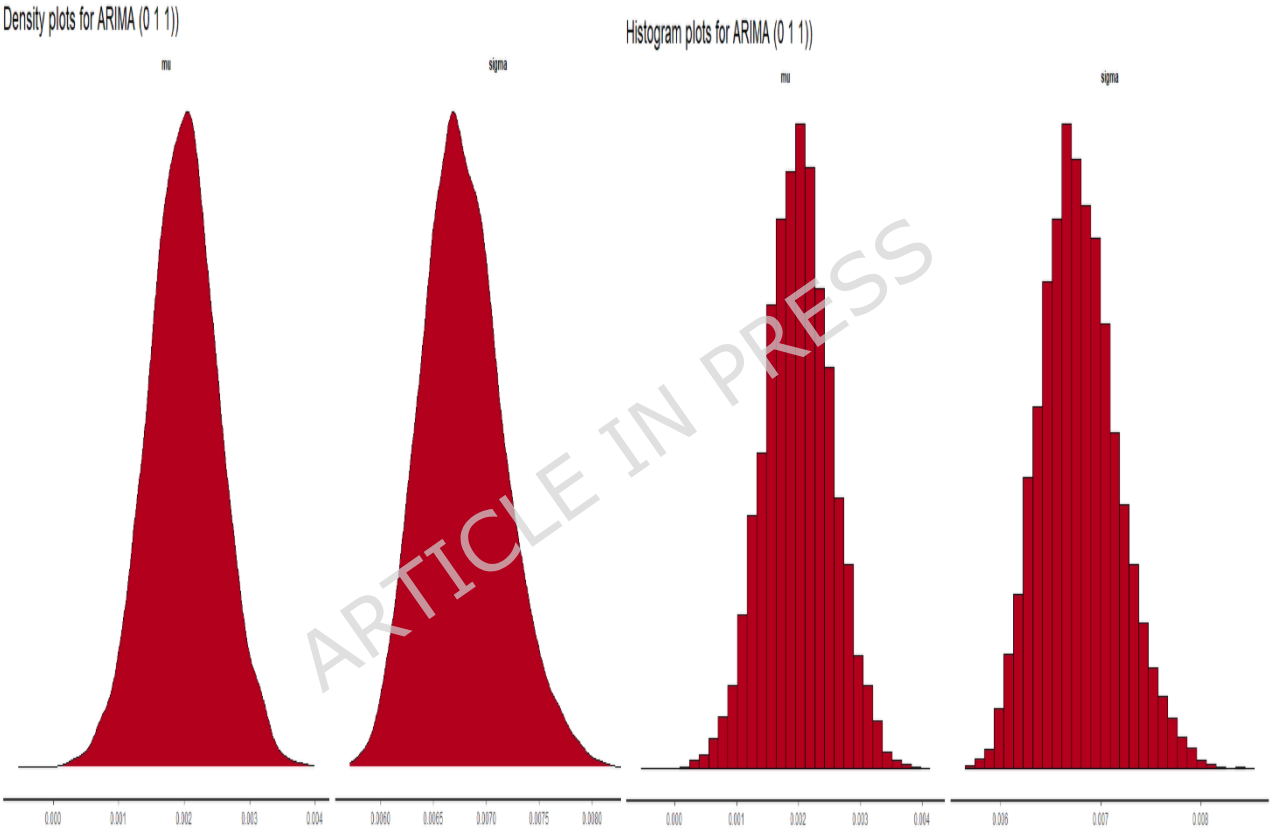


Figure 6: Posterior density and histogram plots for the Bayesian ARIMA (0, 1, 1) model parameters.

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Figure 7 displays the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of the first-differenced CO_2 emissions series. Following first differencing, the series achieved stationarity as

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confirmed by the Augmented Dickey–Fuller (ADF) test. The ACF and PACF plots show only limited significant autocorrelation patterns, indicating that the non-stationary trend present in the original series was successfully removed and supporting the identification of appropriate ARIMA model components.

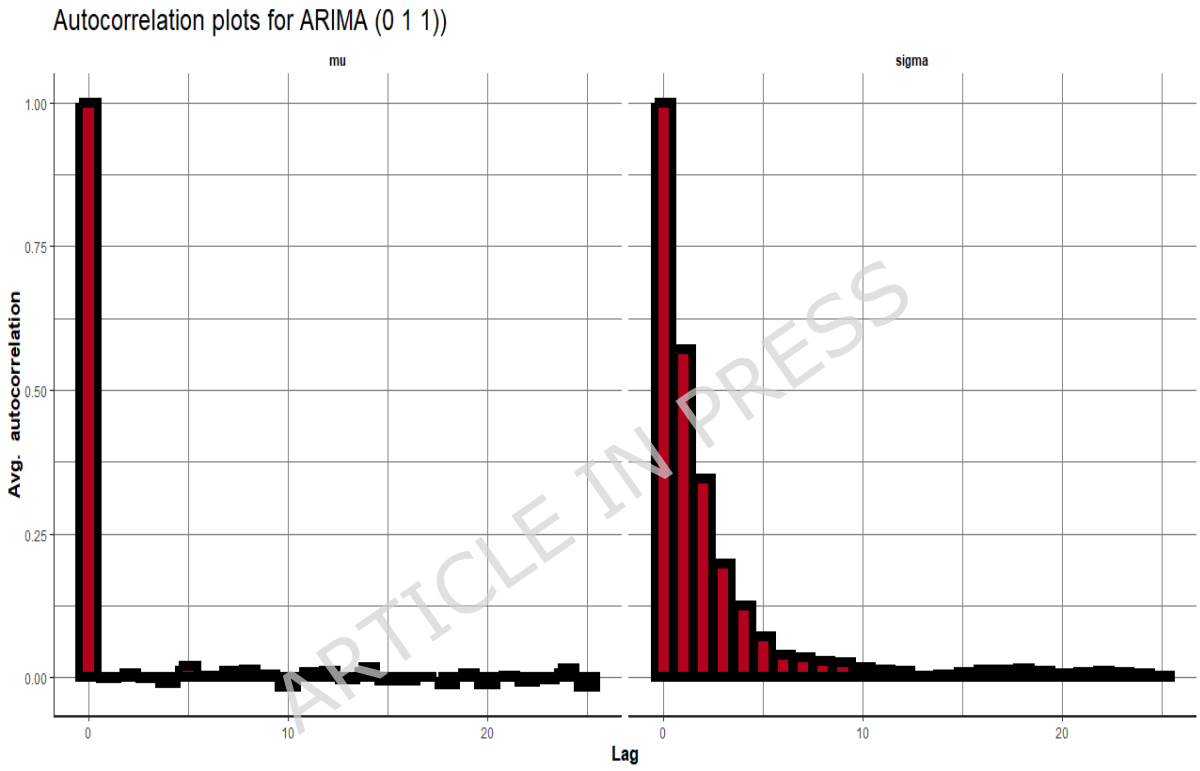


Figure 7: Autocorrelation plots of the posterior distributions for the Bayesian ARIMA (0, 1, 1) model parameters.

4.4. Bayesian Structural Time Series

In this study, the Bayesian Structural Time Series (BSTS) model was implemented using a Stan-based state-space formulation to model Ethiopia's annual CO_2 emissions. The BSTS framework decomposes the observed time series into latent components, including observation error, local level, and slope processes, allowing flexible modeling of trend dynamics and uncertainty. Weakly informative prior distributions were assigned to the

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model parameters to improve model identifiability and stabilize posterior estimation. Specifically, normal priors with mean 0 and variance 1 were assigned to the error terms (μ_err and δ_err). For the variance-related parameters, weakly informative priors were specified for σ_slope, σ_level, and σ_obs to ensure positivity and avoid overly restrictive assumptions while allowing the data to primarily inform posterior inference.

Posterior estimation was conducted using Markov Chain Monte Carlo (MCMC) simulation. The posterior summaries for the main BSTS parameters (s_obs, s_level, and s_slope) are presented in Table 7, including posterior means, standard deviations, 95% credible intervals, effective sample sizes (n_eff), and Gelman–Rubin convergence diagnostics ($\hat{R}$).Table 7 indicates relatively small posterior variance estimates across all three parameters, with mean values of 0.002 for s_obs, 0.006 for s_level, and 0.001 for s_slope, suggesting low variability in observation noise, level evolution, and slope dynamics. The 95% credible intervals are relatively narrow, indicating moderate posterior precision. However, the convergence diagnostics reveal limitations in posterior sampling. The effective sample sizes were relatively low for several parameters, particularly s_obs (n_eff = 26), indicating inefficient mixing of the Markov chains. In addition, the Gelman–Rubin diagnostic values exceeded the recommended threshold of 1.01 for all parameters, with the highest value observed for s_obs ($\hat{R}$ = 1.154), followed by s_level ($\hat{R}$ = 1.053) and s_slope ($\hat{R}$ = 1.032). These results suggest incomplete convergence of the MCMC chains, particularly for the observation variance parameter.

To improve model convergence, additional MCMC iterations, extended warm-up periods, and alternative tuning settings were tested. Although these modifications improved posterior stability, minor convergence issues persisted for certain parameters. This is likely attributable to the relatively complex state-space structure of the BSTS model combined with the limited

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information content of the long-term historical CO_2 emissions data. Therefore, while the BSTS model provides useful insights into the latent trend structure and uncertainty in Ethiopia's CO_2 emissions, its results should be interpreted with caution. These convergence limitations were explicitly considered in the comparative model evaluation and contributed to the Bayesian ARIMA model being selected as the preferred forecasting approach due to superior convergence diagnostics and lower WAIC and LOOIC values.

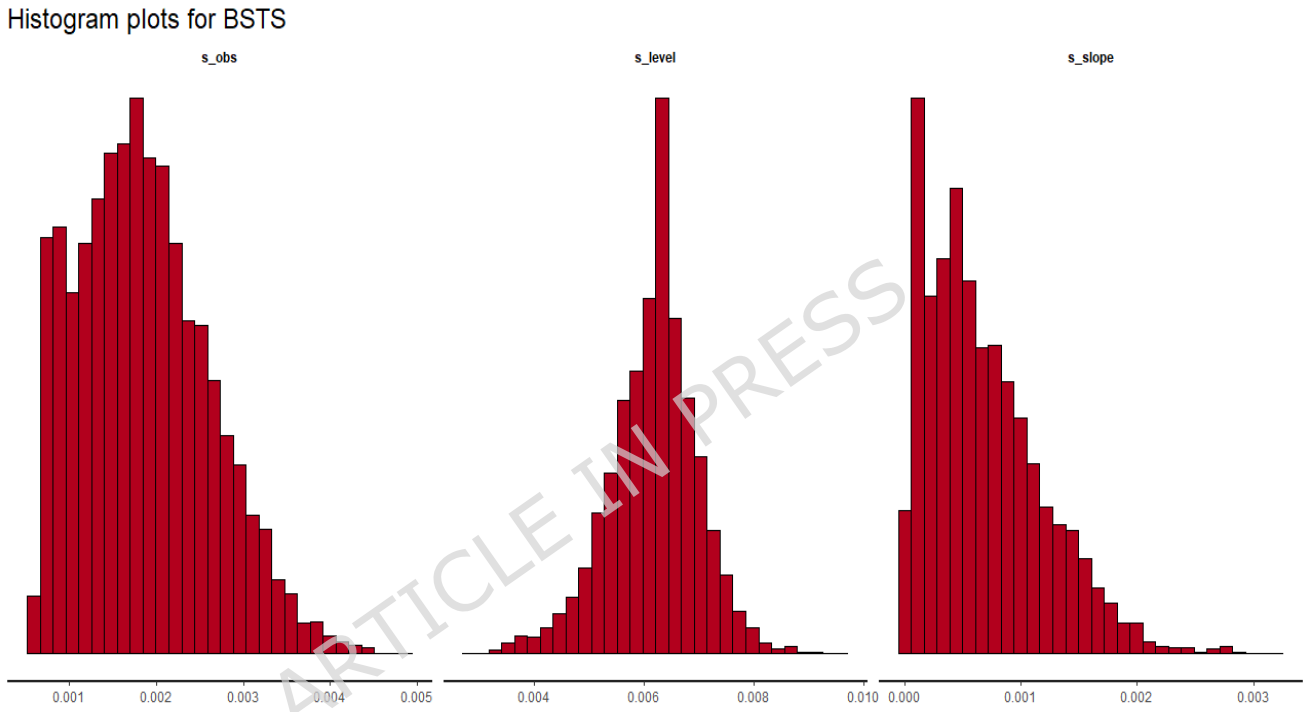
Table 7: **Posterior Summary of Bayesian Structural Time Series Parameters**

Parameters	**Mean**	**Se_mean**	**Sd**	**2.5%**	**50%**	**97.5%**	**n_eff**	**Rhat**
s_obs	0.002	0	0.001	0.001	0.002	0.003	26	1.154
s_level	0.006	0	0.001	0.004	0.006	0.008	87	1.053
s_slope	0.001	0	0.001	0.000	0.001	0.002	75	1.032

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Figures 8, 9, 10and 11 display the histogram plots of the posterior distribution for the BSTS model, the caterpillar plots of the posterior distribution, the density plots of the posterior distribution and the trace plots of the posterior distribution for the BSTS model, respectively. The trace plot in Figure 11 demonstrates the convergence of the Markov chain, albeit limited by space. The trace plot depicted in Figure 11 shows how the MCMC method approaches the target posterior distribution. The caterpillar plot



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shown in Figure 9 clearly reveals that all coefficients lie within the 95% credible intervals and are statistically significant.

Figure 8: Histogram plots illustrating the posterior distributions of the parameters in the BSTS model.

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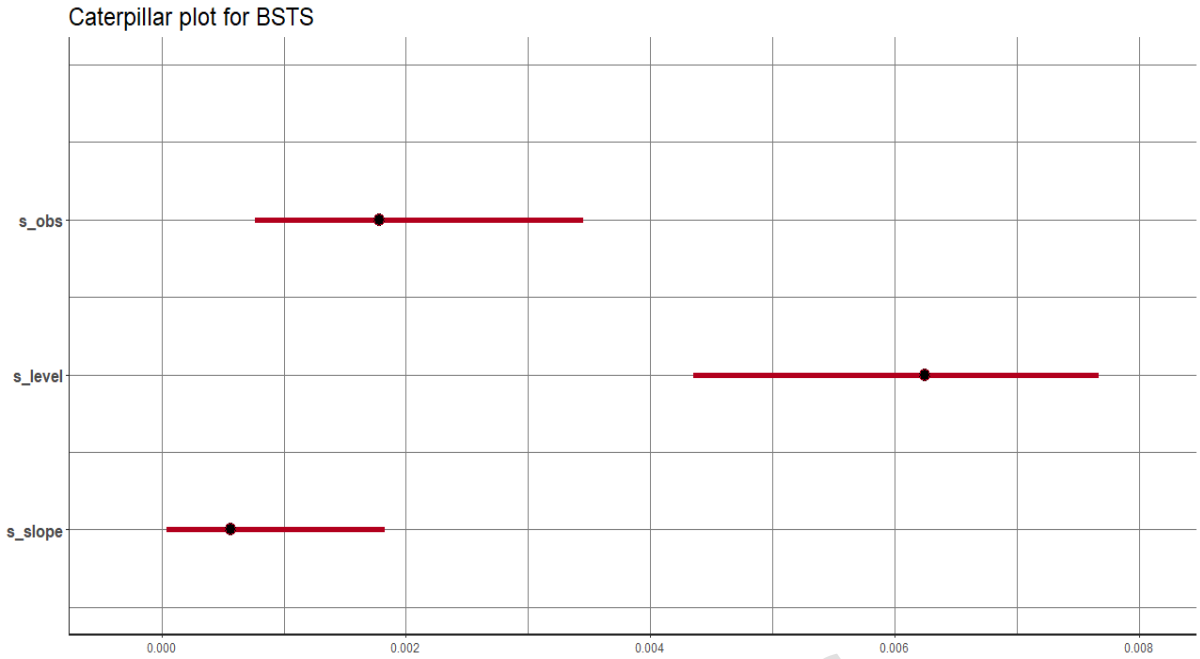
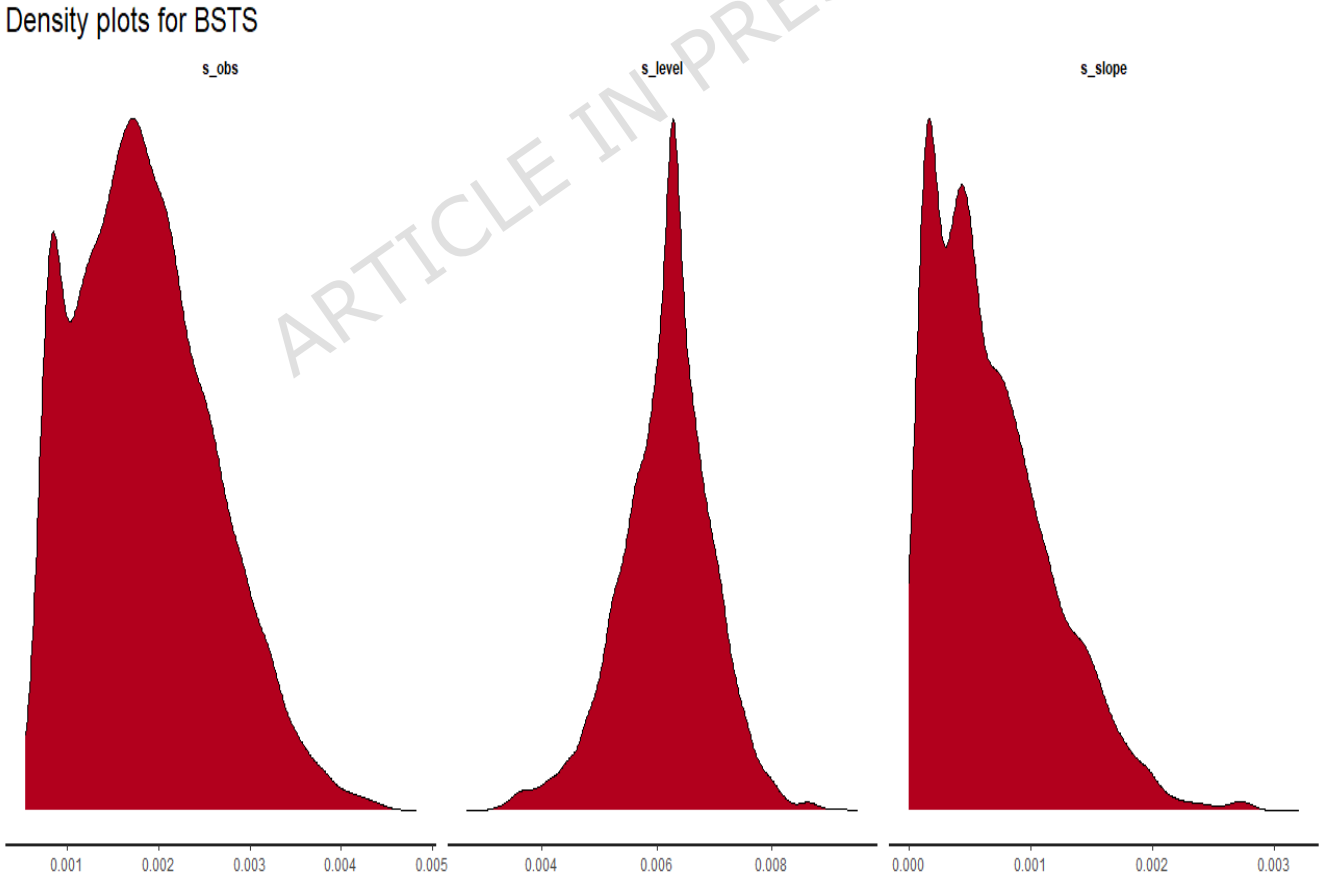


Figure 9: Caterpillar plots of the posterior distributions for the BSTS model parameters.



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Figure 10: Density plots illustrating the posterior distributions of the BSTS model parameters.

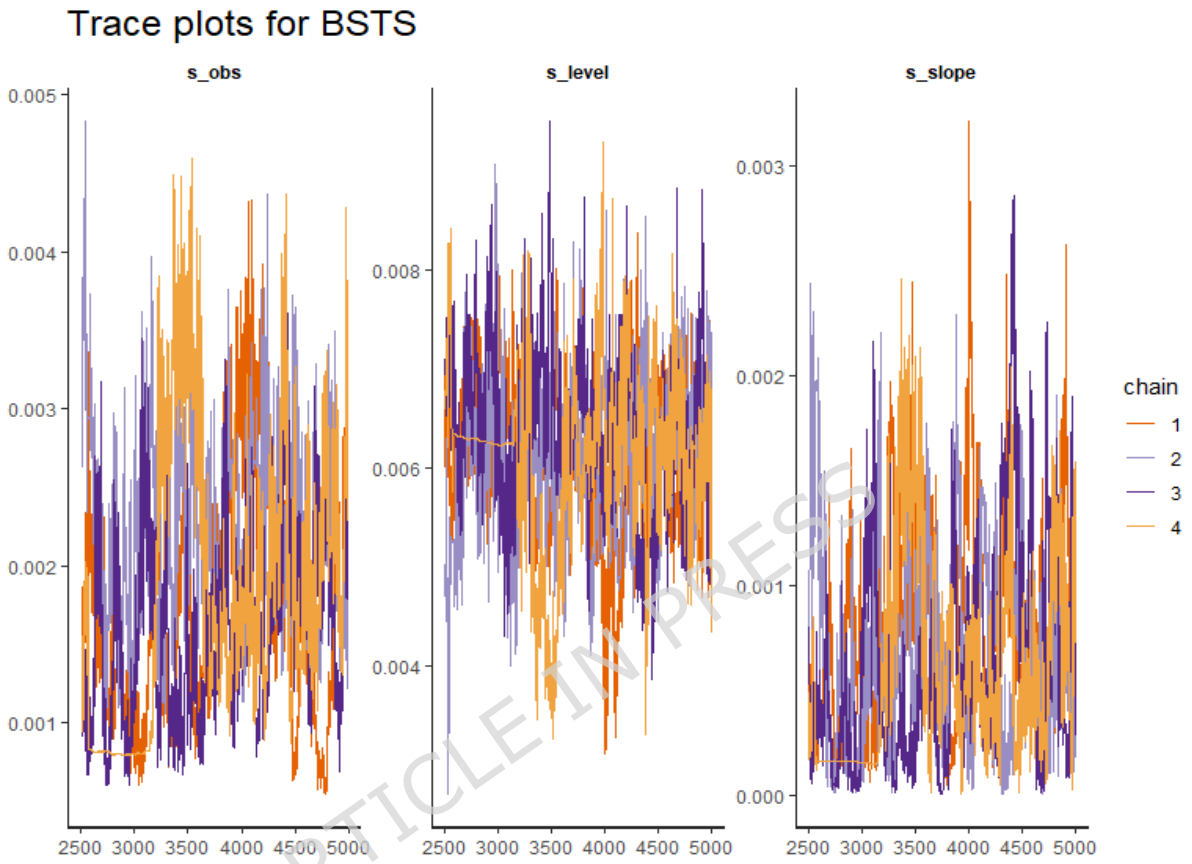


Figure 11: Trace plots showing the convergence of the posterior distributions for the BSTS model.

After analyzing the time series decomposition of historical carbon dioxide emissions data in Ethiopia, a clear trend emerges. To capture this trend component, multiple BSTS models will be developed, including the local level, local linear trend, semi-local linear trend, and Student Local Linear Trend models. Each model will be iterated using MCMC with two different numbers of iterations: 500 and 1000. The performance of these BSTS models will be evaluated using the R-squared value to determine the most suitable model for predicting carbon dioxide emissions per capita in Ethiopia. As shown in Table 8, the R-squared values across the models are very similar, ranging

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from 99.39% to 99.78%. The highest R-squared value of 99.78% is achieved by the BSTS model incorporating the Student Local Linear Trend components with 500 MCMC iterations. Given this outstanding R-squared value, it can be concluded that this model effectively predicts carbon dioxide emissions per capita in Ethiopia.

Table 8: Comparison of R-Square for Carbon Dioxide Emissions per capita in Ethiopia

State Component	Iteration MCMC	R-Square	Residuals.sd	Prediction.sd
Local level trend	n=500	0.9965771	0.002326474	0.006842435
	n=1000	0.9963485	0.002402898	0.006849482
Semi-local Linear trend	n=500	0.9939234	0.003099766	0.006652803
	n=1000	0.9944663	0.002958063	0.006653411
Local Linear Trend	n=500	0.9968539	0.002230427	0.006877666
	n=1000	0.9959805	0.002521064	0.006964582
Auto-Regressive	n=500	0.9970861	0.002146536	0.006828879
	n=1000	0.9968101	0.002245907	0.006833366
Student Local Linear Trend	n=500	**0.9978353**	0.001850129	0.006760423
	n=1000	0.9967854	0.002254552	0.006787838

The lower panel of Figure 12 displays the original carbon dioxide emissions, while the upper panel presents the cumulative absolute errors. The endpoints of the lines in the upper panel indicate the percentage of the mean absolute prediction error among the competing models. Consistent with previous observations, the Local Level and Semi-local Linear Trend models show the lowest cumulative predictive errors. Figure 14 illustrates the forecasted point (FP) for the BSTS model over the next decade as a blue line, with the surrounding green line representing the 95% credible interval around the forecast.

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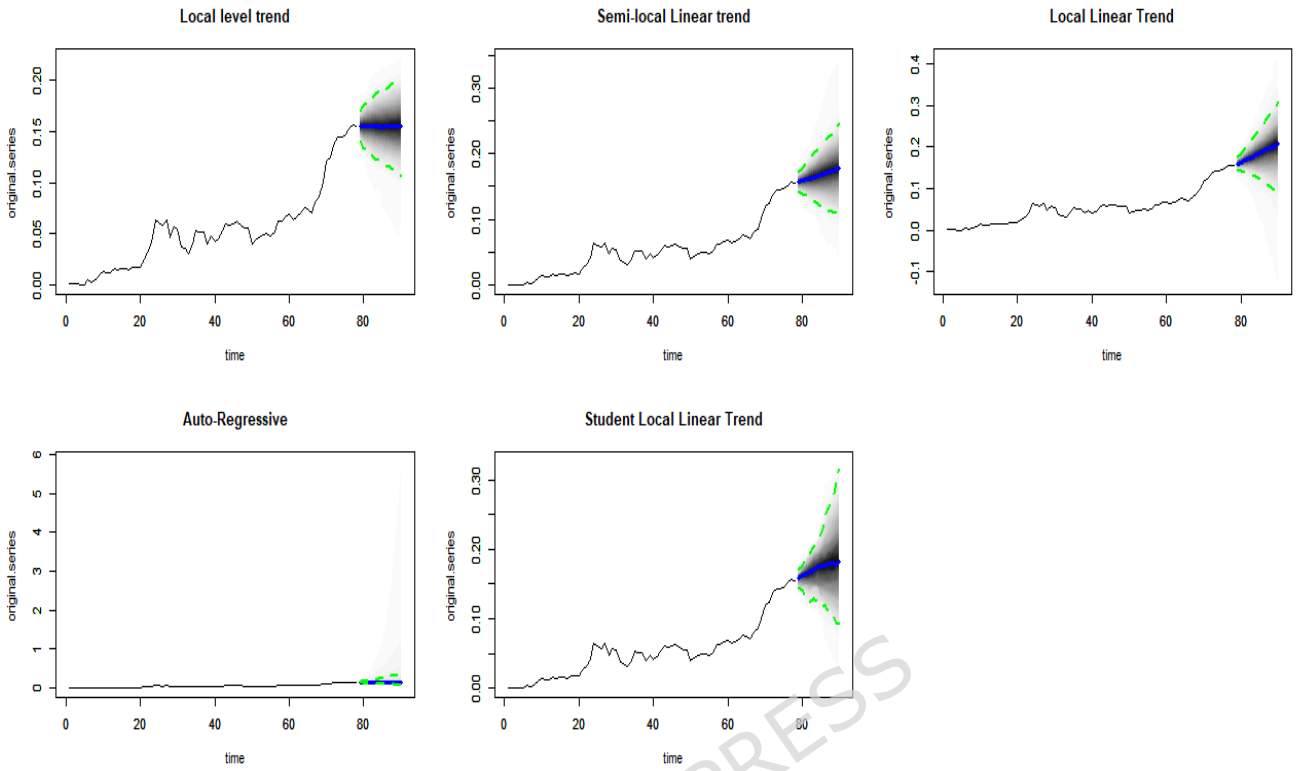


Figure 12: BSTS model components and their future predictions of carbon dioxide emissions.

3.5. Models Comparison of Bayesian ARIMA and BSTS

This study examined two models for time series analysis, specifically the Bayesian ARIMA and BSTS models as shown in Table 9. The process of selecting the ARIMA model incorporates both LOOIC and WAIC metrics. It is now essential to evaluate these models to assess their effectiveness and appropriateness for the analysis. We have considered the leave-one-out cross-validation (LOOIC) criterion in our comparison of the fitted models. A lower value from the selection method indicates a better model fit. LOOIC is one of the fully Bayesian selection methods compared with other information criteria. This technique, which uses the log-likelihood derived from posterior parameter simulations, provides a measure of point-wise out-of-sample prediction accuracy for a fitted Bayesian model. In essence, the Bayesian ARIMA (0 1 1) model emerges as the most suitable for data concerning carbon

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dioxide emissions in Ethiopia and is likely the best choice for the given dataset see Table 9.

Table 9: Comparison of accuracy measures between the Bayesian ARIMA (0,1,1) and BSTS models.

Model	**MPE**	**MAPE**	**RMSE**
Bayesian ARIMA(0 1 1)	$5.074e^{-03}$	$6.846e^{-02}$	0.0366
BSTS model	-26.239	26.45722	0.2872

4.6. Forecasting of Carbon Dioxide Emissions in Ethiopia

A primary objective of this study was to develop an accurate and reliable model for forecasting future CO_2 emissions in Ethiopia. Following parameter estimation, model diagnostics, and comparative evaluation, the Bayesian ARIMA (0,1,1) model was identified as the best-performing model for forecasting Ethiopia's CO_2 emissions. This selection was based on superior predictive performance, stronger convergence diagnostics, and lower values of the Leave-One-Out Information Criterion (LOOIC) and Widely Applicable Information Criterion (WAIC) compared with the BSTS model.

Table 10 presents the forecasted annual per capita carbon dioxide (CO_2) emissions in Ethiopia generated using the Bayesian ARIMA (0,1,1) model for the period 2019–2030, together with 80% and 95% credible intervals. The results indicate an overall increasing trend in Ethiopia's per capita CO_2 emissions over the forecast period. Specifically, forecasted emissions rise from 0.1546 metric tons per capita in 2019 to 0.1785 metric tons per capita by 2030, representing a gradual long-term increase in annual emissions intensity. This upward trend is consistent with the historical pattern observed in Ethiopia's emissions data and suggests continued growth in carbon emissions over time.

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Between 2019 and 2023, forecasted emissions increase moderately from 0.1546 to 0.1650 metric tons per capita, reflecting a continuation of recent historical growth patterns. From 2024 onward, the model projects further increases, reaching 0.1666, 0.1686, 0.1709, 0.1705, 0.1751, 0.1772, and 0.1785 metric tons per capita for the years 2024 through 2030, respectively. Although a slight decline is observed between 2026 and 2027 (from 0.1709 to 0.1705), the general pattern remains upward, indicating minor short-term fluctuations within an overall increasing trajectory. The forecast intervals provide important information regarding uncertainty in future emissions estimates. For example, the 95% credible interval widens from 0.1410–0.1683 in 2019 to 0.1329–0.2251 in 2030. Similarly, the 80% credible interval expands from 0.1458–0.1636 in 2019 to 0.1489–0.2091 in 2030. This widening of both 80% and 95% credible intervals over time reflects increasing uncertainty as the forecasting horizon extends, which is expected in time series forecasting. Ethiopia's per capita emissions remain comparatively low, increasing from approximately 0.1546 metric tons per capita in 2019 to 0.1785 metric tons per capita by 2030. This level is considerably lower than global averages, which stood around 4.8 metric tons per capita in 2022. The relatively low emissions are consistent with Ethiopia's economic structure and heavy reliance on biomass energy. For context, the European Union (28 member states) achieved a 25.7% reduction in CO_2 emissions by 2014 relative to 1990 levels, as part of commitments under the Kyoto Protocol [24, 52, 53]. These contrasting trends highlight Ethiopia's ongoing development and emphasize the importance of adopting sustainable policies to manage anticipated emission increases[57, 58].

Despite the projected increase, Ethiopia's forecasted per capita CO_2 emissions remain relatively low compared with the global average. This is likely attributable to the country's current economic structure, relatively low level of industrialization, and continued reliance on biomass-based energy systems. Nevertheless, the upward trajectory may reflect the effects of rapid

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urbanization, industrial expansion, infrastructure development, increasing transportation demand, and rising energy consumption associated with economic growth. Similar trends have been reported in developing economies such as Malaysia, Bangladesh, Pakistan, and South Africa, where increasing emissions have been linked to urbanization, industrialization, and higher energy demand[18–21, 23, 25, 59]. These findings suggest that Ethiopia may follow a comparable emissions pathway as structural transformation continues. Moreover, recent studies have emphasized the importance of probabilistic approaches for analyzing environmental sustainability and carbon dynamics under uncertainty [57, 58], supporting the application of Bayesian methods in carbon emissions forecasting.

The projected increase in emissions through 2030 has important implications for carbon management and climate policy in Ethiopia. Continued growth in CO_2 emissions underscores the need to strengthen the transition toward low-carbon development pathways. Specifically, the findings suggest the importance of expanding renewable energy infrastructure, promoting cleaner transportation systems, improving industrial energy efficiency, and reinforcing the implementation of Ethiopia's Climate-Resilient Green Economy strategy and other sustainable development initiatives [55, 56], [55,56]. These recommendations are directly motivated by the forecasted upward trend and are consistent with previous studies highlighting the importance of integrating environmental sustainability with economic development [57–60].

Beyond the Ethiopian context, this study contributes to the broader literature on carbon balance and management by demonstrating the applicability of Bayesian time-series approaches for environmental forecasting under uncertainty. The comparative analysis showed that the Bayesian ARIMA (0,1,1) model outperformed the BSTS model in terms of predictive accuracy, convergence diagnostics, and information criteria. Moreover, the Bayesian

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framework provides probabilistic forecasts and credible intervals, thereby enabling a more comprehensive assessment of uncertainty and supporting evidence-based climate policy and long-term environmental planning. These methodological insights may also be valuable for other developing countries facing similar challenges related to limited data availability and increasing pressure to achieve sustainable low-carbon growth. The combined ARIMA-BSTS model used in this study integrates linear ARIMA components with conditional variance modelling from BSTS.

The parameter optimization for the ARIMA (0 1 1)-BSTS (1) model was conducted using the maximum likelihood approach, employing the Marquardt (1963) and Berndt-Hall-Hall-Hausman algorithms. Three distributions, namely normal, t-student, and generalized error, were considered for this estimation. Additionally, the covariance coefficient was estimated using the reversed matrix by Hessian[18, 54, 60].

Table 10: Forecasted Annual Per Capita Carbon Dioxide Emissions in Ethiopia (2019–2030)

Years	**Forecasted Carbon Dioxide Emission of Bayesian ARIMA(0 1 1)**	**Lo 80**	**Hi 80**	**Lo 95%**	**Hi 95%**
2019	0.1546	0.1457559	0.1635981	0.1410334	0.1683206
2020	0.1586	0.1452354	0.1681186	0.1341199	0.1752341
2021	0.1605	0.1459409	0.1760591	0.1379691	0.1840309
2022	0.1631	0.1456112	0.1803888	0.1364062	0.1895938
2023	0.1650	0.1455587	0.1844413	0.1352672	0.1947328
2024	0.1666	0.1457032	0.1882968	0.1344293	0.1995707
2025	0.1686	0.1459968	0.1920032	0.1338196	0.2041804

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2026	0.1709	0.1464085	0.1955915	0.1333906	0.2086094
2027	0.1705	0.1469168	0.1990832	0.1331092	0.2128908
2028	0.1751	0.1475059	0.2024941	0.1329514	0.2170486
2029	0.1772	0.1481640	0.2058360	0.1328991	0.2211009
2030	0.1785	0.1488817	0.2091183	0.1329381	0.2250619

Table 11 presents the forecast accuracy measures for the Bayesian ARIMA model. The model demonstrates relatively good predictive performance, with low values of Mean Absolute Error **(**MAE = 0.0047) and Root Mean Square Error **(**RMSE = 0.0067**),** indicating small average forecast errors. The residual autocorrelation at lag 1 **(**ACF1 = 0.081**)** is close to zero, suggesting that residuals are approximately uncorrelated and that the model has adequately captured the temporal dependence structure in the data. The Mean Absolute Scaled Error **(**MASE = 0.961**)** is slightly below 1, indicating that the Bayesian ARIMA model performs marginally better than a naïve benchmark forecast.

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However, both the Mean Absolute Percentage Error **(**MAPE = 0**)** and Mean Percentage Error **(**MPE = 0**)** should be interpreted cautiously. Because Ethiopia's historical CO_2 emissions per capita contain very small values, particularly in the earlier years of the time series, percentage-based error measures become unstable and may produce misleading or degenerate values. Therefore, greater emphasis was placed on scale-dependent accuracy measures such as MAE, RMSE, and MASE for evaluating forecasting performance. Overall, the results suggest that the Bayesian ARIMA model provides reasonably accurate forecasts and performs slightly better than the naïve benchmark model.

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Table 11: Accuracy measurements for the Bayesian ARIMA (0,1,0) model.

ACF1	MAE	MAPE	MASE	MPE	ME	RMSE
0.08147572	0.004743603	0	0.9612037	0	-1.28e-08	0.006696727

The yearly variations in per capita carbon dioxide emission rates in Ethiopia are projected to increase steadily from 2019 to 2030, as illustrated in Figures 13 and 14. This upward trend reflects the persistent pattern observed in the historical data. However, it is important to note that the model forecasts assume continuity of existing trends and do not explicitly account for potential structural breaks, policy changes, or economic shocks that could affect future emissions. Therefore, while the graph highlights a clear pattern in Ethiopia's CO_2 emissions, these projections should be interpreted as baseline estimates subject to change with evolving socio-economic or environmental factors. This observation further motivates a thorough assessment of the stationarity of the data series to ensure appropriate model selection and forecasting reliability.

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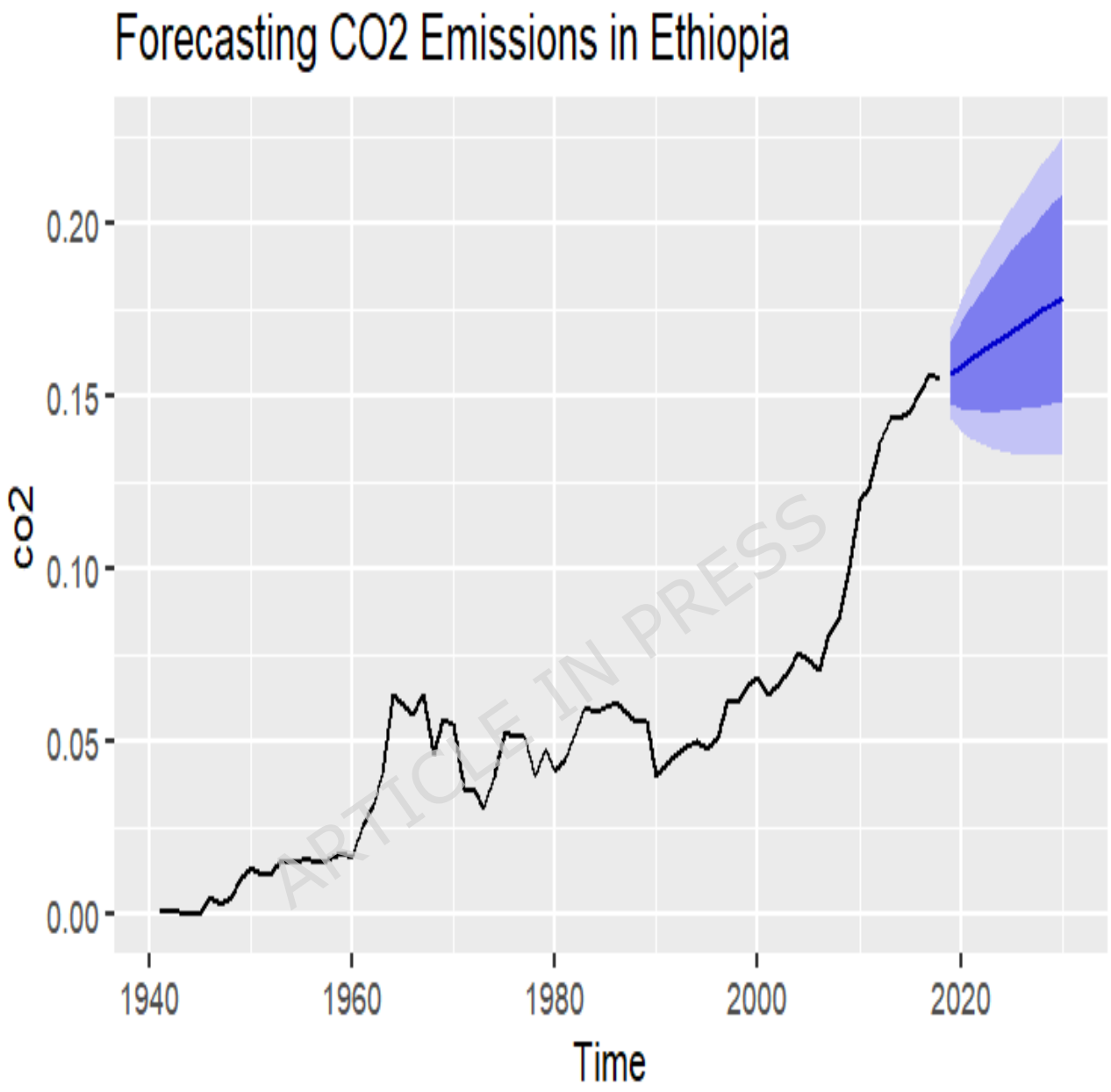


Figure 13: Forecast plot of per capita carbon dioxide emissions in Ethiopia from 2019 to 2030

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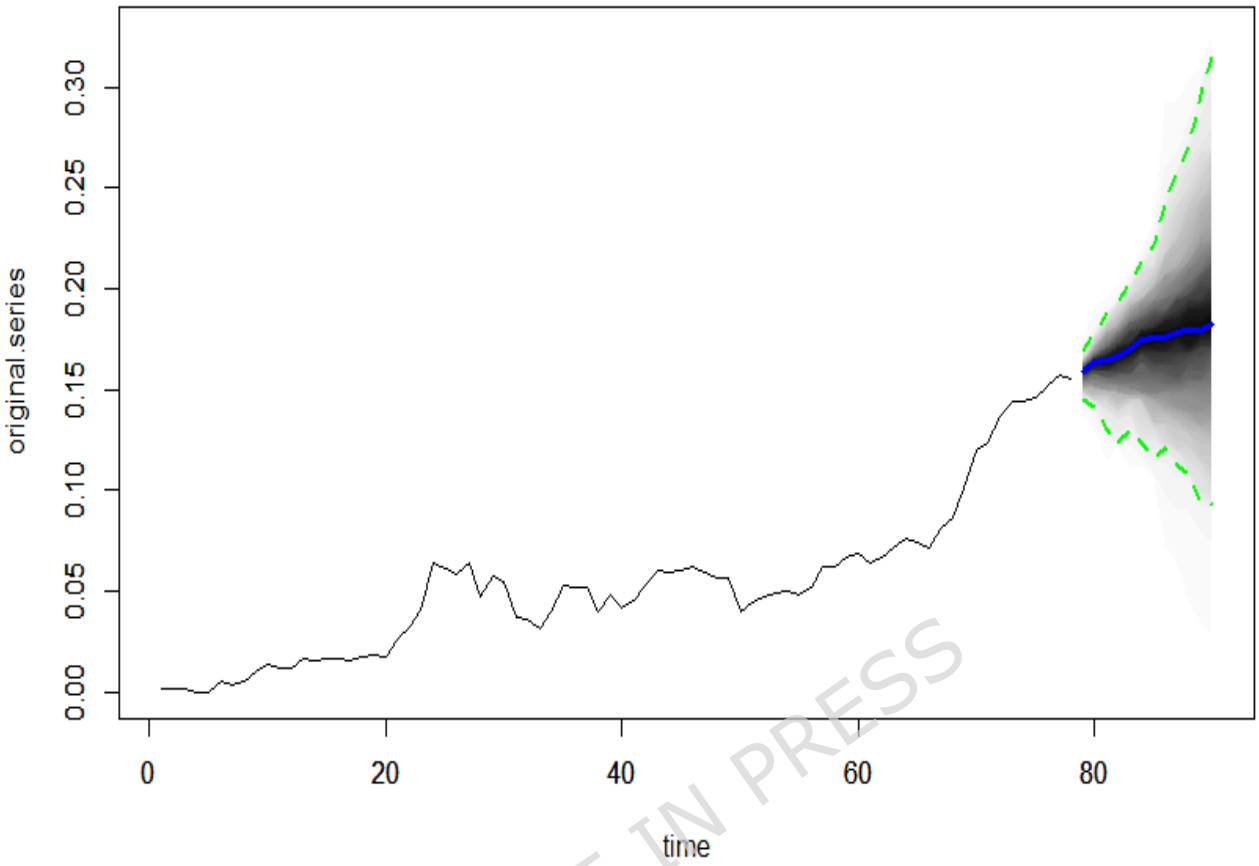


Figure 14: BSTS model forecast of carbon dioxide emissions in Ethiopia with 95% credible intervals.

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Limitations of the Study

This study is limited by the use of only historical annual CO_2 emissions data without incorporating key exogenous variables such as energy use, industrialization, urbanization, and policy factors. In addition, formal out-of-sample validation methods such as rolling forecasts or cross-validation were not conducted due to limited annual time series data. Finally, the forecasts assume continuity of historical trends and do not account for future structural breaks, policy changes, technological innovations, or economic shocks that may affect Ethiopia's CO_2 emissions.

5. Conclusions and Recommendations

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This study analyzed Ethiopia's per capita carbon dioxide (CO_2) emissions from 1941 to 2022 using Bayesian time series models, specifically comparing Bayesian Autoregressive Integrated Moving Average (ARIMA) and Bayesian Structural Time Series (BSTS) approaches. The results indicate that the Bayesian ARIMA (0, 1,1) model provided the best overall forecasting performance based on information criteria and diagnostic results, although minor convergence limitations were observed in the BSTS model.

Forecasts indicate that Ethiopia's per capita CO_2 emissions are expected to increase by approximately 15.4% between 2019 and 2030, signaling a significant environmental challenge that requires urgent, data-driven policy intervention. To address this projected growth, Ethiopia should set specific emission reduction targets, aiming for an average annual decrease or offset of about 1.3% in per capita emissions to align with its Nationally Determined Contributions (NDCs). The forecasting results suggest a gradual increase in Ethiopia's per capita CO_2 emissions over the projection period, indicating a persistent upward trend associated with ongoing structural changes in the economy, including urbanization, industrial expansion, transport growth, and rising energy demand. While the level of emissions remains relatively low in global terms, the projected trajectory highlights the importance of proactive planning to avoid future escalation of emissions.

Based on these findings, several policy implications can be drawn. Strengthening the transition toward renewable energy sources such as solar, wind, and hydropower is essential to reduce reliance on biomass and fossil fuels. Improving energy efficiency across industrial, transport, and household sectors can further help moderate emission growth. In addition, integrating low-carbon development strategies into national planning frameworks, including Ethiopia's Climate Resilient Green Economy strategy, is important for ensuring sustainable long-term development. Finally, future research should extend the current analysis by incorporating multivariate frameworks

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that include key socioeconomic and environmental drivers such as GDP, energy consumption, urbanization, and population growth. This would improve both the explanatory power and policy relevance of forecasting models and provide a more comprehensive understanding of the determinants of CO_2 emissions in Ethiopia.

Disclosure

Authors declared that there are no conflicts of interest in this work.

Ethics Approval and Consent to Participate

This study is based on secondary data concerning carbon dioxide (CO_2) emissions, sourced from the publicly available World Bank database (https://databank.worldbank.org/databases). The data do not involve human participants or animal subjects. Therefore, ethical approval and consent to participate were not required. However, the research protocol was reviewed and approved by the Ethical Approval Committee of Jigjiga University.

Authors' contributions

Buzuneh Tasfa Marine, Abdisalan Ahmed Osman, and Dagne Tesfaye Mengistie contributed to the conceptualization of the study, methodology development, and implementation of the Bayesian ARIMA and BSTS models for CO_2 emissions forecasting in Ethiopia. They conducted data pre-processing, statistical analysis, model estimation, and results interpretation. They also contributed to manuscript drafting and revision. Buzuneh Tasfa Marine is the corresponding author and is responsible for manuscript submission, communication with the journal, handling revisions, and coordinating the overall research process during manuscript development and revision stages. Daud Hussein Adawe and Rediat Takele contributed to data collection, conceptual support, and provided critical feedback throughout the writing process.

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Availability of Data

This study used secondary data obtained from international emission databases. The carbon dioxide (CO_2) emission data are publicly available from the World Bank's World Development Indicators and can be accessed through the World Bank data portal.(data base link at: http://data.worldbank.org/indicator/EN.ATM.CO2E.PC?location=BD).

Consent for publication

Not applicable.

Competing interests

Authors declared that there is no conflict of interest between the authors.

Reference

1. Kabir, M., et al., Climate change due to increasing concentration of carbon dioxide and its impacts on environment in 21st century; a mini review. Journal of King Saud University-Science, 2023. 35(5): p. 102693.
2. Amoo, L.M. and R.L. Fagbenle, Climate change in developing nations of the world. Applications of Heat, Mass and Fluid Boundary Layers, 2020: p. 437-471.
3. Olivier, J.G., K. Schure, and J. Peters, Trends in global CO2 and total greenhouse gas emissions. PBL Netherlands Environmental Assessment Agency, 2017. 5(1).
4. Aralica, Z. and N. Stojcic, Regional patterns of deindustrialization and prospects for reindustrialization in South and Central East European countries. 2015, wiiw Balkan Observatory Working Papers.

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ARTICLE IN PRESS

5. Vally, S. and C.A. Spreen, Human rights in the World Bank 2020 education strategy, in The World Bank and education: Critiques and alternatives. 2012, SensePublishers Rotterdam. p. 173-187.
6. You, W., Y. Zhang, and C.-C. Lee, The dynamic impact of economic growth and economic complexity on CO2 emissions: An advanced panel data estimation. Economic Analysis and Policy, 2022. 73: p. 112-128.
7. Dritsaki, M. and C. Dritsaki, Forecasting European Union CO2 emissions using autoregressive integrated moving average-autoregressive conditional heteroscedasticity models. International Journal of Energy Economics and Policy, 2020. 10(4): p. 411-423.
8. Belay, B., D. Diriba, and F. Senbeta, Estimation of GHGs emission from traditional kilns charcoal production in northwestern Ethiopia: Implications on climate change. Heliyon, 2024. 10(24).
9. Engdaw, B.D., Assessment of the trends of greenhouse gas emission in Ethiopia. Geography, Environment, Sustainability, 2020. 13(2): p. 135-146.
10. Rehman, M.Z., A.A. Dar, and T. Wangmo A, Forecasting CO2 Emissions in India: A Time Series Analysis Using ARIMA. Processes, 2024. 12(12): p. 2699.
11. Almarashi, A.M. and K. Khan, Bayesian structural time series. Nanoscience and Nanotechnology Letters, 2020. 12(1): p. 54-61.
12. Box, G. and G. Jenkins, Analysis: Forecasting and Control. San francisco, 1976.
13. Navas Thorakkattle, M., S. Farhin, and A.A. Khan, Forecasting the trends of covid-19 and causal impact of vaccines using bayesian structural time series and arima. Annals of Data Science, 2022. 9(5): p. 1025-1047.
14. Kalinina, I. and A. Gozhyj, Modeling and forecasting of nonlinear nonstationary processes based on the Bayesian structural time series. Applied Aspects of Information Technology, 2022. 3(5): p. 240-255.

ARTICLE IN PRESS

ARTICLE IN PRESS

15. Rodríguez, Y. and O.D. Olariaga, Air traffic demand forecasting with a Bayesian structural time series approach. Periodica Polytechnica Transportation Engineering, 2024. 52(1): p. 75-85.
16. Abu Bakar, N. and S. Rosbi, Autoregressive integrated moving average (ARIMA) model for forecasting cryptocurrency exchange rate in high volatility environment: A new insight of bitcoin transaction. International Journal of Advanced Engineering Research and Science, 2017. 4(11): p. 130-137.
17. Grubb, M., et al., A review of Chinese CO2 emission projections to 2030: the role of economic structure and policy. Climate Policy, 2015. 15(sup1): p. S7-S39.
18. Costantini, L., et al., Forecasting national CO2 emissions worldwide. Scientific Reports, 2024. 14(1): p. 22438.
19. Engo, J., Driving forces and decoupling indicators for carbon emissions from the industrial sector in Egypt, Morocco, Algeria, and Tunisia. Environmental Science and Pollution Research, 2021. 28(12): p. 14329-14342.
20. Baloch, M.A., N. Mahmood, and J.W. Zhang, Effect of natural resources, renewable energy and economic development on CO2 emissions in BRICS countries. Science of the total environment, 2019. 678: p. 632-638.
21. Raihan, A., et al., Nexus between carbon emissions, economic growth, renewable energy use, urbanization, industrialization, technological innovation, and forest area towards achieving environmental sustainability in Bangladesh. Energy and Climate Change, 2022. 3: p. 100080.
22. Iftikhar, H., et al., Modeling and forecasting carbon dioxide emission in Pakistan using a hybrid combination of regression and time series models. Heliyon, 2024. 10(13).
23. Mustaffa, A.S. and A. Shabri. An improved rolling NGBM (1, 1) forecasting model with GRG nonlinear method of optimization for fossil carbon

ARTICLE IN PRESS

dioxide emissions in Malaysia and Singapore. in 2020 11th IEEE control and system graduate research colloquium (ICSGRC). 2020. IEEE.

24. Zhou, X., et al., Study on CO2 Emission Forecast of “Four Provinces of Mountains and Rivers” Based on Time-SeriesMachine Learning. Atmosphere, 2024. 15(8): p. 949.
25. Sulaiman, A.S.M. and A. Shabri, Forecasting carbon dioxide emissions for Malaysia using grey model with Cramer’s rule. Mal J Fund Appl Sci, 2021: p. 437-445.
26. Monica, C., et al., CO2 emissions of all world countries. 2022.
27. Stephens, I.E., et al., 2022 roadmap on low temperature electrochemical CO2 reduction. Journal of Physics: Energy, 2022. 4(4): p. 042003.
28. Ma, C., et al., Biomass derived nitrogen and sulfur co-doped porous carbons for efficient CO2 adsorption. Separation and Purification Technology, 2022. 281: p. 119899.
29. Jin, S., et al., Advances and challenges for the electrochemical reduction of CO2 to CO: from fundamentals to industrialization. Angewandte Chemie, 2021. 133(38): p. 20795-20816.
30. Cui, C., et al., Evolution-based CO2 emission baseline scenarios of Chinese cities in 2025. Applied Energy, 2021. 281: p. 116116.
31. Pole, A., M. West, and J. Harrison, Applied Bayesian forecasting and time series analysis. 2018: Chapman and Hall/CRC.
32. Pang, X. and J. Gill, Spike and slab prior distributions for simultaneous Bayesian hypothesis testing, model selection, and prediction, of nonlinear outcomes. Unpublished manuscript, 2009.
33. Egeh, O.M., C. Chesneau, and A.H. Muse, Exploring hybrid models for forecasting CO 2 emissions in drought-prone Somalia: a comparative analysis. Earth Science Informatics, 2023. 16(4): p. 3895-3912.
34. Zellner, A., Statistical analysis of econometric models. Journal of the American Statistical 2026. doi:10.1007/s40974-026-00420-z. Association, 1979. 74(367): p. 628-643.

ARTICLE IN PRESS

35. Georgescu I, Kinnunen J. Bayesian assessment of ecological footprint drivers in Finland: a model averaging approach under structural and model uncertainty. *Energy, Ecology and Environment*. 2026. doi:10.1007/s40974-026-00420-z.

36. Manigandan P, Pachiyappan D, Radhamma J, Sha N. Forecasting of carbon emissions in India using Bayesian ARIMA and BSTS approaches: evidence from environmental sustainability. *Nature Environment and Pollution Technology*. 2026. doi:10.46488/NEPT.2026.v25i02.001.

37. Wang Z, Li YP, Huang GH, Gong JW, Li YF, Zhang Q. A factorial-analysis-based Bayesian neural network method for quantifying China's CO_2 emissions under dual-carbon target. *Science of the Total Environment*. 2024;920:170698. doi:10.1016/j.scitotenv.2024.170698.

38. Bayesian spatio-temporal modeling of African CO_2 emissions (1990–2020): A hierarchical approach to identify determinants, regional trends, and local dynamics. *Science of the Total Environment*. 2025;969:178894. doi:10.1016/j.scitotenv.2025.178894.

39. Powell, A.J., et al., Predicting future disease burden in a rapidly changing climate. 2020, Sandia National Lab.(SNL-NM), Albuquerque, NM (United States); Sandia ….

40. Abbink, J., Religion in public spaces: Emerging Muslim–Christian polemics in Ethiopia. African Affairs, 2011. 110(439): p. 253-274.

41. De Siqueira, A.C., The 2020 World Bank education strategy: Nothing new, or the same old gospel, in The World Bank and Education. 2012, Brill. p. 67-81.

42. Dixit, A. and S. Jain, Contemporary approaches to analyze non-stationary time-series: Some solutions and challenges. Recent Advances in Computer Science and Communications (Formerly: Recent Patents on Computer Science), 2023. 16(2): p. 61-80.

ARTICLE IN PRESS

ARTICLE IN PRESS

43. Schaffer, A.L., T.A. Dobbins, and S.-A. Pearson, Interrupted time series analysis using autoregressive integrated moving average (ARIMA) models: a guide for evaluating large-scale health interventions. BMC medical research methodology, 2021. 21(1): p. 58.
44. Vehtari, A., A. Gelman, and J. Gabry, Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. Statistics and computing, 2017. 27(5): p. 1413-1432.
45. Luo, Y. and K. Al-Harbi, Performances of LOO and WAIC as IRT model selection methods. Psychological Test and Assessment Modeling, 2017. 59(2): p. 183.
46. Shumway, C.A., The evolution of complex brains and behaviors in African cichlid fishes. Current zoology, 2010. 56(1): p. 144-156.
47. Nelson, C.R. and C.R. Plosser, Trends and random walks in macroeconmic time series: some evidence and implications. Journal of monetary economics, 1982. 10(2): p. 139-162.
48. Tibebe, D., et al., Homogenous Climatic Regions for Targeting Green Water Management Technologies in the Abbay Basin, Ethiopia. Climate, 2023. 11(10): p. 212.
49. Scott, S.L., Bayesian methods for Hidden Markov Models: Recursive computing in the 21st century. Journal of the American statistical Association, 2002. 97(457): p. 337-351.
50. Rahman, A. and M.M. Hasan, Modeling and forecasting of carbon dioxide emissions in Bangladesh using Autoregressive Integrated Moving Average (ARIMA) models. Open Journal of Statistics, 2017. 7(4): p. 560-566.
51. Lartillot, N., Identifying the best approximating model in Bayesian phylogenetics: Bayes factors, cross-validation or wAIC? Systematic Biology, 2023. 72(3): p. 616-638.
52. Qiao, Z., X. Meng, and L. Wu, Forecasting carbon dioxide emissions in APEC member countries by a new cumulative grey model. Ecological Indicators, 2021. 125: p. 107593.

ARTICLE IN PRESS

53. Dhamija, A.K. and V.K. Bhalla, Financial time series forecasting: comparison of neural networks and ARCH models. International Research Journal of Finance and Economics, 2010. 49: p. 185-202.

54. Ajala, A.A., et al., An examination of daily CO2 emissions prediction through a comparative analysis of machine learning, deep learning, and statistical models. Environmental Science and Pollution Research, 2025. 32(5): p. 2510-2535.

55. Zhou, C., Zhang, W., Richardson-Barlow, C. *et al.* Navigating carbon neutrality: policy pathways and consistency on industrial decarbonization in China. *Carbon Balance Manage* **20**, 66 (2025). https://doi.org/10.1186/s13021-025-00356-7

56. Wang, C., Wang, F., Zhang, H., Ye, Y., Wu, Q., & Su, Y. (2014). Carbon Emissions Decomposition and Environmental Mitigation Policy Recommendations for Sustainable Development in Shandong Province. *Sustainability*, *6*(11), 8164-8179. https://doi.org/10.3390/su6118164

57. Tian, Y., Ren, X., Li, K., & Li, X. (2025). Carbon Dioxide Emission Forecast: A Review of Existing Models and Future Challenges. *Sustainability*, *17*(4), 1471. https://doi.org/10.3390/su17041471

58. Georgescu I, Kinnunen J. Bayesian assessment of ecological footprint drivers in Finland: a model averaging approach under structural and model uncertainty. Energy, Ecology and Environment. 2026 Apr 27:1-27.

59. Manigandan P, Pachiyappan D, Radhamma J, Sha N. Forecasting of Carbon Emissions in India Using Bayesian ARIMA and BSTS Approaches: Evidence from Environmental Sustainability.

60. Afrifa-Yamoah E, Osei PM. Bayesian spatio-temporal modeling of African CO2 emissions (1990–2020): A hierarchical approach to identify

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determinants, regional trends, and local dynamics. Science of the Total Environment. 2025 Mar 15;969:178894.

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